Competitive Exam Practice 2: Computer-System Organization and Architecture

GATE (Computer Science) Pattern

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Concepts That Click

Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern — the repo holds no verifiable GATE OS past-year source.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about control transfer to the kernel is **TRUE**?

(A) An interrupt is synchronous with the instruction that raised it, whereas a trap is asynchronous.

(B) A divide-by-zero fault is delivered as a hardware interrupt, because the arithmetic unit signals it.

(C) A system call enters the kernel through a trap, whereas a disk-completion signal is delivered as an interrupt.

(D) Both interrupts and traps can occur only at an instruction boundary chosen by the running process.

Q2 *[Original, GATE-pattern]* **[NAT, 2 marks]** An interrupt vector table starts at physical address 4096, and entry i occupies the 8 bytes beginning at address $4096 + 8i$. The CPU receives interrupt number 40. The physical address at which the CPU fetches this interrupt's handler address is _____.

Q3 *[Original, GATE-pattern]* **[NAT, 2 marks]** A network adapter raises 4,000 interrupts per second. Saving and restoring the context and dispatching to the handler costs 4.5 μs per interrupt, and the device handler itself runs for 8 μs. Ignoring all other overhead, the percentage of CPU time consumed by this adapter's interrupts is _____ %.

Q4 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about maskable interrupts and priority is **TRUE**?

(A) Clearing the CPU's interrupt-enable flag also prevents delivery of non-maskable interrupts.

(B) A masked interrupt request is discarded, so it can never be serviced at a later time.

(C) While a high-priority handler runs with lower-priority interrupts masked, a pending lower-priority request is normally serviced after the high-priority handler returns.

(D) Interrupt priority is determined solely by the order in which handlers are installed in software; the interrupt controller treats all requests equally.

Q5 *[Original, GATE-pattern]* **[NAT, 2 marks]** A magnetic disk rotates at 7,200 RPM. Its average seek time is 6 ms, its sustained transfer rate is 128 MB/s, and one block is 4 KB. The average time to read one block, in milliseconds rounded to two decimal places, is _____.

Q6 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about volatility in the memory hierarchy is **TRUE**?

(A) DRAM is non-volatile, which is why a program's variables survive a power failure.

(B) SRAM caches are non-volatile, whereas DRAM is volatile.

(C) Registers and DRAM lose their contents without power, whereas magnetic disk and NAND flash retain theirs.

(D) NAND flash is volatile, but it has a lower cost per byte than magnetic disk.

Q7 *[Original, GATE-pattern]* **[NAT, 2 marks]** A two-level cache hierarchy has L1 hit time 1 ns and L1 hit rate 90%. After an L1 miss is detected, an L2 access costs 8 ns, and 80% of L1 misses hit in L2. An access that misses in both caches costs a further 60 ns for main memory. The average memory access time (AMAT), in nanoseconds, is _____.

Q8 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about polling, interrupt-driven I/O, and DMA is **TRUE**?

(A) Polling never reads the device's status register, so it cannot determine whether the device is ready.

(B) In interrupt-driven I/O the device signals the CPU when service is needed, so CPU cost is incurred per event rather than by busy-waiting.

(C) DMA removes all cache-coherence concerns, because the controller writes directly into main memory.

(D) DMA is cheaper than a single interrupt for a one-byte transfer from a slow keyboard, since setup cost is negligible.

Q9 *[Original, GATE-pattern]* **[NAT, 2 marks]** A 2 GHz CPU spends 2,000 cycles programming a DMA controller and 1,000 cycles in the completion interrupt handler. The controller then moves 1 MB at 200 MB/s, a transfer lasting 5 ms. The DMA-related CPU time, expressed as a percentage of this 5 ms transfer time and rounded to two decimal places, is _____ %.

Q10 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about multiprocessor, multicore, and cluster architectures is **TRUE**?

(A) In a multicore processor, all cores share a single set of architectural registers.

(B) SMP requires every processor to have private main memory; otherwise the processors cannot run one OS image.

(C) A cluster connects independent computers over a network and can continue service after one whole machine fails.

(D) A cluster is a single chip containing several cores that share a last-level cache.

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